

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 3.3 EXCRETION AND HOMEOSTASIS IN ANIMALS: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How do different animals maintain a balanced internal environment and eliminate toxic wastes despite drastic changes in their external conditions? Think back to the hot afternoon in a Kenyan community: a dog panting with its tongue hanging out, a lizard sitting still on a warm rock, goats that drink water yet make concentrated urine, and freshwater fish that keep excreting large amounts of dilute urine. Answer each section in the blank space under the prompt. Use evidence from what you observed in Lessons 1 to 6 (charts, the goat or cow kidney dissection, the nephron animations, the liver and skin photomicrographs, the feedback flowcharts and the thermoregulation notes). Write in full sentences and use correct biology words. Where a section asks for a claim, evidence and reasoning, give all three. At the end you will use your answers to help build your annotated poster model of how a mammal regulates its internal environment and excretes waste, which your teacher will mark with a rubric.

Section 1: Getting rid of wastes in different habitats

Insects, birds, freshwater fish and mammals such as goats make different nitrogenous wastes and use different excretory structures. (a) Name the nitrogenous wastes ammonia, urea and uric acid and state which animals in our charts excrete each one. (b) Name the excretory structures used by insects, birds, fish and mammals. (c) Explain how the type of waste an animal makes is linked to how much water is available to it. Use the freshwater fish and the bird or insect as your two examples.

Write your answer here:

Section 2: From blood to urine in the goat

On the hot afternoon the goat drank water yet produced concentrated urine. Using what you learned from the goat or cow kidney dissection and the nephron animations: (a) Describe the roles of the cortex, medulla and pelvis. (b) Explain, in order, how ultrafiltration, selective reabsorption and secretion in the nephron turn blood into urine. (c) Explain why glucose is absent from urine but urea is concentrated in it.

Write your answer here:

Section 3: Liver and skin as partners of the kidney

Urea is found in the goat's urine and a little urea is found in sweat. (a) Explain where the urea comes from and how it reaches the kidney. (b) State two other jobs of the liver and what the skin excretes. (c) A resource person spoke about cirrhosis, hepatitis and fatty liver disease. Explain why damage to the liver would be a problem for the whole body.

Write your answer here:

Section 4: Negative feedback in water, glucose and temperature

(a) Explain what negative feedback means. (b) Describe the feedback loop that adjusts how much water the goat's kidney reabsorbs on a hot day, and explain why the freshwater fish produces dilute urine. (c) Describe how insulin and glucagon keep blood glucose steady after a meal of ugali and sukuma wiki. (d) Choose diabetes mellitus or diabetes insipidus and explain which part of its loop has failed, using its symptoms.

Write your answer here:

Section 5: Final explanation of the hot afternoon

Write a claim-evidence-reasoning explanation that answers the driving question using the dog, the lizard, the goat and the freshwater fish. Include: (a) how the dog and the lizard each keep a steady body temperature, and the difference between them; (b) how the goat and the freshwater fish handle water and wastes differently; (c) why excretion and homeostasis are essential for survival. Then state one idea from your Lesson 1 model that you have changed.

Write your answer here:

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Accuracy of excretion science (wastes, kidney, liver, skin)	Correctly names ammonia, urea and uric acid with the animals and structures, and accurately explains ultrafiltration, selective reabsorption, secretion and the roles of the liver and skin, linking waste type to water availability.	Mostly correct explanation of wastes, nephron steps and the roles of the liver and skin, with minor gaps or one weak link to water availability.	Several errors or missing parts; confuses wastes, structures or nephron steps, or lists facts without explaining.
Explanation of homeostasis and negative feedback	Clearly explains negative feedback and applies it correctly to ADH and water, insulin and glucagon and temperature control, including the role of the hypothalamus and effectors and the difference between the dog and the lizard.	Explains negative feedback and applies it to most examples with small omissions.	Describes changes without showing a loop, or confuses the responses of endotherms and ectotherms.
Use of evidence and claim-	Gives a clear claim supported by specific	Gives a claim with some evidence from	Claim is unclear or evidence is vague,

evidence-reasoning	evidence from the lessons (dog, lizard, goat, freshwater fish, dissection, case studies) and reasoning that links evidence to the claim.	the lessons, but the reasoning is partly missing.	missing or not from the lessons.
Answering the driving question and reflecting on learning	Fully connects all four animals to the driving question, explains why excretion and homeostasis are essential for cell survival, and states a specific change from the Lesson 1 model.	Connects most animals to the driving question and gives a general reason and a simple change in thinking.	Leaves out animals or gives no reason why excretion and homeostasis matter; no clear reflection.
Scientific vocabulary and clarity	Uses scientific terms correctly and consistently in well-organised, clear sentences.	Uses most terms correctly; writing is mostly clear.	Uses few scientific terms or uses them incorrectly; writing is hard to follow.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.